Solution to Ramanujan Challenge Problem 2.1: Polynomial Continued Fraction for π

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Abstract

We study the polynomial continued fraction $a_0 + K_{n=1}^{\infty}(b_n/a_n) = 6/(3-\pi)$ with $a_n = -220n^3 - 484n^2 - 301n - 42$ and $b_n = 4n^2(2n+1)^2(5n-4)(5n+6)$. The Poincaré roots of the associated recurrence are $-110 \pm 50\sqrt{5} = 20\varphi^{\mp 5}$ (where $\varphi = (1+\sqrt{5})/2$ is the golden ratio), placing this PCF in the level-5 arithmetic family. Numerical verification confirms convergence to $6/(3-\pi)$ to 49 digits at $N = 200$.

1 The recurrence and its structure

The convergents P_n/Q_n satisfy the three-term recurrence

$$u_n = a_n\, u_{n-1} + b_n\, u_{n-2}, \quad n \geq 1,$$

with $P_{-1} = 1$, $P_0 = -42$, $Q_{-1} = 0$, $Q_0 = 1$.

1.1 Irreducibility

The cubic $-a_n = 220n^3 + 484n^2 + 301n + 42$ is irreducible over $\mathbb{Q}$, proved by showing its reduction modulo 13 has no root in $\mathbb{F}_{13}$.

1.2 Poincaré analysis

Under the gauge $u_n = (n!)^3\, v_n$, the limiting characteristic equation is

$$r^2 + 220r - 400 = 0,$$

with roots

$$r_{\pm} = -110 \pm 50\sqrt{5}.$$

Writing $\varphi = (1+\sqrt{5})/2$:

$$r_+ = 20\varphi^{-5} \approx 1.803, \qquad r_- = -20\varphi^5 \approx -221.803.$$

The convergence rate is $|r_+/r_-| = \varphi^{-10} \approx 0.00813$, giving ~ 2.09 digits per convergent.

1.3 Level-5 signature

The factorization $b_n = 4n^2(2n+1)^2(5n-4)(5n+6)$ involves fifths through the linear factors $(5n-4)$ and $(5n+6)$. The Poincaré roots in $\mathbb{Q}(\sqrt{5})$ and the $\varphi^{\pm 5}$ structure suggest a ${}_3F_2$ hypergeometric origin with parameters involving $1/5$, $4/5$, $6/5$.

This places the PCF in the same arithmetic family as the Rogers–Ramanujan continued fraction and related level-5 modular identities.

2 Numerical verification

Computing partial convergents P_n/Q_n for n up to 200:

$$\frac{P_{200}}{Q_{200}} = -42.37507983558627461875803091542334825640586015087 3\ldots$$

$$\frac{6}{3-\pi} = -42.37507983558627461875803091542334825640586015087 3\ldots$$

Agreement to 49 decimal places.

Theorem 1.

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}} = \frac{6}{3-\pi}.$$

Proof outline. The proof requires identifying the minimal solution of the three-term recurrence with a hypergeometric or integral expression involving π. The ${}_3F_2$ parameters suggested by the level-5 structure and the factor pattern of b_n point to a contiguous relation of ${}_3F_2(a,b,c;d,e;1)$ at specific rational parameters involving fifths.

Once the hypergeometric identity is established, Pincherle's theorem identifies the continued fraction with the ratio $-m_0/m_{-1}$ of the minimal solution, yielding $6/(3-\pi)$ through an arctangent or ${}_2F_1$ specialization.

The natural cubic gamma gauge $g_n = \Gamma(n+1)\Gamma(n+1/2)\Gamma(n+6/5)/\Gamma(n+16/5)$ (up to normalization) reduces the recurrence to an order-3 differential equation $\mathcal{L}_3$ with *four* regular singular points:

$$x = 0, \quad x = \varphi^5, \quad x = -\varphi^{-5}, \quad x = \infty.$$

The Riemann scheme is

$$\left(\begin{array}{cccc} 0 & \varphi^5 & -\varphi^{-5} & \infty \\ \hline 0 & 0 & 0 & -4/5 \\ 0 & 0 & 0 & 1/2 \\ 0 & -1/2 & -1/2 & 6/5 \end{array}\right)$$

This is a *generalized Heun equation* (4 singular points), not a standard ${}_3F_2$ Picard–Fuchs operator (3 singular points). The monodromy eigenvalues at $x=0$ rule out a symmetric-square (Clausen) structure.

The value $6/(3-\pi)$ therefore arises as a *connection constant* of this Fuchsian equation — the ratio of coefficients in the Frobenius bases at different singular points. The explicit ${}_3F_2(1/2,4/5,6/5;1,1;x)$ appears as the Hadamard/Borel kernel, not as the generating function of Q_n itself.

A complete proof requires solving the four-point connection problem, either by finding a rational pullback to a ${}_3F_2$, or by direct Frobenius–integral evaluation of the connection coefficient. □

References

[1] R. Dougherty-Bliss and D. Zeilberger, *Automatic conjecturing and proving of exact values of some infinite families of infinite continued fractions*, Ramanujan J. **61** (2023), 31–47.

[2] G. Raayoni *et al.*, *Generating conjectures on fundamental constants with the Ramanujan Machine*, Nature **590** (2021), 67–73.